

Matthew P. Grindle, PhD

Physical Chemist | NMR Specialist | Enhanced Oil Recovery Researcher

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PROFESSIONAL SUMMARY

Physical chemist and nuclear magnetic resonance (NMR) specialist applying low-field NMR spectroscopy and magnetic resonance imaging to enhanced oil recovery (EOR) and subsurface fluid-rock interactions. As an ORISE postdoctoral fellow at the National Energy Technology Laboratory, develops experimental and analytical methods for high-pressure studies of CO₂, natural gas, water, and surfactant injection in unconventional reservoir rocks, including shales, carbonates, and caprocks. Experienced in quantifying multiphase fluid saturation, pore structure, wettability, and transport behavior, with a record of cross-disciplinary research, conference publication, mentoring, and laboratory instruction.

CORE EXPERTISE

Low-field NMR and MRI | T1/T2 relaxometry | CO₂ huff-and-puff EOR | Shale characterization | High-pressure core analysis | Multiphase fluid saturation | NMR/EPR spectroscopy | Python, MATLAB, Linux, Bash, Microsoft Office

EDUCATION

2025 **PhD, Chemistry** | Miami University | Oxford, OH

Dissertation: Structural Dynamics of Inorganic Complexes and Chaperone Proteins Investigated Through NMR Spectroscopy and Molecular Dynamics

2016 **BS, Chemistry** | Purdue University | West Lafayette, IN

PROFESSIONAL EXPERIENCE

2025-Present **Postdoctoral Fellow** | National Energy Technology Laboratory (NETL) | US DOE | Pittsburgh, PA

- Serve as the NMR subject-matter expert for petroleum, geologic materials, and subsurface-energy research.
- Develop low-field NMR and MRI methods to quantify porosity, pore-size distributions, fluid saturation, wettability, and multiphase fluid movement in unconventional reservoir samples.
- Design and conduct high-pressure enhanced oil recovery experiments using CO₂, natural gas, water, and surfactants, with emphasis on cyclic huff-and-puff recovery from unconventional reservoir rock.
- Create analysis workflows for relaxation and imaging data to track fluid redistribution, pore-scale transport, and oil recovery throughout injection experiments.
- Collaborate with federal, university, contractor, and industry partner researchers and communicate results through conference and journal publications

2019-2025 **Graduate Research Assistant** | Miami University | Oxford, OH

- Investigated transition-metal complexes and biomolecular systems using NMR, EPR, UV-vis spectroscopy, ICP analysis, X-ray diffraction, and computational modeling.
- Mentored 6 undergraduate and 2 graduate researchers in spectroscopy, physical chemistry, data analysis, and laboratory methods.
- Co-authored journal manuscripts and presented research at 7 scientific conferences.

2019-2024 **Teaching Assistant, General Chemistry** | Miami University | Oxford, OH

- Prepared lectures and laboratory instruction for classes of 20-100 students across five academic years.
- Coordinated meetings and training for 20 teaching assistants as head teaching assistant to support consistent instruction and laboratory safety.

2017-2019 **Fermentation Associate** | DuPont Industrial Biosciences | Wilmington, DE

- Operated and supported 24 pilot fermenters for 14 L scale-up experiments in a continuous research environment.
- Troubleshoot Python-based controls and in-house data-tracking systems and coordinated with polymerization and separations laboratories.

2016-2017 **Analyst** | Bioanalytical Systems, Inc. | West Lafayette, IN

- Conducted GLP pharmaceutical analyses for quality control and stability studies; maintained and troubleshoot 15 Agilent HPLC systems.
- Developed five analytical standard operating procedures for future pharmaceutical studies.

- 2014-2016 Microbiologist** | Tate & Lyle | West Lafayette, IN
- Performed microbiological and environmental monitoring for a corn starch and syrup manufacturing facility, including Listeria surveillance.
- 2014-2015 Student Co-op Chemist** | Chemtura Corporation | West Lafayette, IN
- Characterized bromine-based flame-retardant materials using NMR, HPLC, GPC, Karl Fischer titration, and related analytical methods.
 - Developed TEMPO-radical-based methods for preparing bisphenol-derived flame-retardant candidates.
- 2014 Quality Management Intern** | Cargill | Hammond, IN
- Conducted chemical quality testing and designed experiments evaluating corn-syrup storage facilities and ion-exchange materials.

PUBLICATIONS

- 2026 Enhanced Oil Recovery Effects of Adding Non-Ionic Surfactants During CO₂ Huff-and-Puff.** M. P. Grindle, D. Tapriyal, L. Burrows, and A. Goodman. *Unconventional Resources Technology Conference (URTeC), Houston, TX.*
- 2026 Comparing Huff-N-Puff Oil Recovery in Unconventional Shale Cores with CO₂ Alone Against Low Concentration Solution of 2EH-PO₅-EO₁₀ in CO₂ Using NMR Spectroscopy.** M. P. Grindle, D. Tapriyal, L. C. Burrows, R. Enick, and A. Goodman. *SPE Improved Oil Recovery Conference, Tulsa, Oklahoma.*
- 2024 NMR Observation of Three Interconverting Structures in TpPh₃Me Cobalt(II) Nitrate.** M. P. Grindle, V. Batra, and D. L. Tierney. *European Journal of Inorganic Chemistry.*
- 2023 Field- and Temperature-Dependent Paramagnetic Relaxation Enhancements in Co(II) Trispyrazolylmethanes.** N. D. Kumarage, A. R. Marts, M. P. Grindle, J. C. Kaine, L. A. Crandall, W.-Y. Chen, C. J. Ziegler, and D. L. Tierney. *Inorganic Chemistry*
- 2021 Structural Communication between the E. coli Chaperones DnaK and Hsp90.** M. P. Grindle, B. Carter, J. P. Alao, K. Connors, R. Tehver, and A. N. Kravats. *International Journal of Molecular Sciences*

SELECTED PRESENTATIONS

- Jun 2026 Enhanced Oil Recovery Effects of Adding Non-Ionic Surfactants During CO₂ Huff-and-Puff.** M. P. Grindle, D. Tapriyal, L. Burrows, and A. Goodman. *Unconventional Resources Technology Conference (URTeC), Houston, Texas.*
- Nov 2023 Solution NMR Magnetic Axis Direction of 4/5-Coordinate Co(II) Complexes** | Graduate Research Forum, Miami University, Oxford, OH
- Apr 2023 NMR Relaxation Dynamics of 4/5-Coordinate Co(II) Complexes** | Experimental NMR Conference (ENC), Pacific Grove, CA
- Mar 2023 Solution Dynamics of 4/5-Coordinate Cobalt Complexes** | American Chemical Society, Indianapolis, IN
- Nov 2022 Physical Electronic Dynamics in Five-Coordinate Cobalt(II)** | Ohio Inorganic Weekend, The Ohio State University, Columbus, OH
- Nov 2022 Solution Dynamics of Co-Based Single-Ion Magnets** | Graduate Research Forum, Miami University, Oxford, OH
- Jan 2021 Computational Modeling of the Modulation of Hsp70 by Hsp90** | Midwest Stress Response and Molecular Chaperones, Virtual
- Nov 2020 Modeling the Modulation of Hsp90 by Hsp70** | Graduate Research Forum, Miami University, Oxford, OH

AWARDS AND LEADERSHIP

- 2026 National Laboratory Research SLAM Finalist** | National Energy Technology Laboratory, US Department of Energy
- 2022-2024 Graduate Student Advocate** | Miami University Department of Chemistry and Biochemistry
- 2022-2023 Outstanding Teaching Assistant** | Miami University Department of Chemistry and Biochemistry
- 2022 Three Minute Thesis Finalist** | Miami University Graduate School

SCIENCE OUTREACH

- Feb 2026** **Western Pennsylvania Science Bowl** | Question Judge | National Energy Technology Laboratory
- Ensured correct pronunciation for Science Bowl participants
 - Kept time for competition
- May 2024** **International Day of Light** | Presenter | Cincinnati Museum Center
- Led chromatography experiments for ~100 parents and students for a fun day of science exploration
- Jun 2022-23** **Talawanda Science Week** | Presenter | Miami University
- Delivered hands-on science experiments for elementary school children
 - Presented hour-long sessions to 20 students, 4 times a day
- Jun 2022-23** **Grandparents College** | Presenter | Miami University
- Taught introductory forensic science experiments to elementary and middle school students accompanied by their grandparents

TECHNICAL SKILLS

- Magnetic resonance: low-field NMR, MRI, NMR relaxometry, EPR spectroscopy, pulse-sequence and data-analysis method development.
- Analytical methods: UV-vis spectroscopy, ICP analysis, X-ray diffraction, HPLC, GPC, and Karl Fischer titration.
- Computational and data tools: Python, MATLAB, Linux, Bash, molecular dynamics, and electronic-structure calculations.